

XAI-Based Intelligent Disease Prediction System with Transparent Decision Insights

A PROJECT REPORT

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ABSTRACT

Epilepsy is one of the most prevalent chronic neurological disorders, affecting approximately 50 million individuals worldwide. Timely and accurate detection of epileptic seizures from electroencephalogram (EEG) signals is vital for patient safety, quality of life, and clinical management. However, manual interpretation of long-duration EEG recordings is labour-intensive and prone to high inter-rater variability. Moreover, existing deep learning models for seizure detection often lack interpretability, which is a critical requirement for clinical adoption.

This project presents a novel framework called Domain-Adapted Graph Representation Learning (DA-GRL) for automated, robust, and clinically interpretable seizure detection from multi-channel EEG. The proposed architecture integrates three key components: (i) a convolutional temporal feature extractor that learns local spectrotemporal representations from raw EEG windows; (ii) a graph representation learning module that models functional connectivity between EEG electrode pairs using graph convolutions, explicitly encoding the spatial topology of the 10–20 electrode system; and (iii) a domain adaptation module based on a gradient reversal layer (GRL) that enforces patient-invariant feature distributions, thereby mitigating the well-known problem of inter-subject EEG variability.

Experiments were conducted on the publicly available CHB-MIT Pediatric Scalp EEG Database, which contains recordings from 24 paediatric patients with intractable seizures. A comprehensive preprocessing pipeline was applied, including bandpass filtering (0.5–40 Hz), notch filtering (50 Hz), z-score normalisation, Daubechies-4 wavelet decomposition, statistical feature extraction, and Power Spectral Density (PSD) analysis. Class imbalance was addressed using the Synthetic Minority Over-sampling Technique (SMOTE), resulting in a balanced dataset of 11,600 samples (5,800 per class).

Four architectural variants were benchmarked: Spiking Convolutional Transformer (SCT), Graph Representation Learning (GRL), Domain-Adapted SCT (DA-SCT), and the proposed DA-GRL. The DA-GRL model achieved the highest performance across all metrics: accuracy 99.7%, precision 99.3%, recall 100%, F1-score 99.7%, and AUC 1.000. The perfect recall score is clinically significant as it implies zero missed seizure events.

Post-hoc explainability was integrated using SHAP (GradientExplainer) and LIME analy-

ses. The C3-P3 and T8-P8 electrode pairs were identified as the most discriminative channels, consistent with known neurophysiological involvement in focal onset seizures. Temporal SHAP heatmaps localised seizure-sensitive time windows within each 4-second segment, providing clinician-interpretable evidence for every model prediction. The combination of state-of-the-art performance and transparent decision support establishes the DA-GRL framework as a strong candidate for real-world neuro-diagnostic assistance systems.

Keywords: EEG, Epileptic Seizure Detection, Graph Representation Learning, Domain Adaptation, Gradient Reversal Layer, Explainable Artificial Intelligence, SHAP, LIME, CHB-MIT Dataset, Deep Learning.

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
AUC	Area Under the Receiver Operating Characteristic Curve
BN	Batch Normalisation
CHB	Children’s Hospital Boston
CNN	Convolutional Neural Network
DA	Domain Adaptation
DA-GRL	Domain-Adapted Graph Representation Learning
DA-SCT	Domain-Adapted Spiking Convolutional Transformer
db4	Daubechies Wavelet of Order 4
DWT	Discrete Wavelet Transform
EEG	Electroencephalogram / Electroencephalography
EDF	European Data Format
FN	False Negative
FP	False Positive
F1	F1-Score (Harmonic Mean of Precision and Recall)
GAT	Graph Attention Network
GCN	Graph Convolutional Network
GRL	Gradient Reversal Layer (also Graph Representation Learning)
Hz	Hertz
LIME	Local Interpretable Model-Agnostic Explanations
MIT	Massachusetts Institute of Technology
MNE	MNE-Python (EEG/MEG Analysis Library)
PSD	Power Spectral Density
ReLU	Rectified Linear Unit
RMS	Root Mean Square
ROC	Receiver Operating Characteristic
SCT	Spiking Convolutional Transformer

SHAP	SHapley Additive exPlanations
SMOTE	Synthetic Minority Over-sampling Technique

CHAPTER 1

INTRODUCTION

1.1 Overview

Epilepsy is a chronic neurological disorder characterised by recurrent, unprovoked seizures that arise from abnormal, excessive, or synchronous neuronal activity in the brain. It ranks among the most disabling non-communicable neurological diseases globally, with an estimated 50 million individuals affected across all age groups, geographies, and socioeconomic strata. In developing countries, the treatment gap — the proportion of people with active epilepsy who do not receive appropriate treatment — exceeds 75%, underscoring both the clinical importance and the humanitarian urgency of improving seizure management tools.

A single unprovoked seizure can result in serious physical injury, including falls, burns, drowning, and in the worst cases, Sudden Unexpected Death in Epilepsy (SUDEP). The psychosocial impact is equally severe: patients face driving restrictions, employment limitations, and significant social stigma. For paediatric patients in particular, repeated seizures during critical periods of neurodevelopment may lead to long-term cognitive impairment and reduced quality of life.

The fundamental clinical challenge in epilepsy management is twofold. First, the accurate diagnosis of seizure type and localisation of the epileptic focus is a prerequisite for appropriate pharmacological therapy and eligibility for surgical resection. Second, even among patients receiving optimal medical treatment, approximately one-third continue to experience drug-resistant seizures, necessitating continuous monitoring and the development of reliable automated prediction tools.

1.2 The Role of Electroencephalography

The scalp electroencephalogram (EEG) remains the primary diagnostic modality for epilepsy. It provides a non-invasive, millisecond-resolution recording of cortical electrical activity across multiple electrode channels. In a standard clinical EEG, electrodes are placed on the scalp according to the International 10–20 system, which defines anatomically standardised

positions including frontal (Fp, F), central (C), parietal (P), occipital (O), and temporal (T) sites.

During a seizure, the EEG exhibits characteristic patterns including rhythmic spike-wave discharges, high-frequency oscillations, and widespread synchronisation of neuronal activity across electrode channels. These patterns are distinct from the background EEG of a normal brain, which is characterised by lower-amplitude, less-synchronised oscillations. This visual distinctiveness forms the neurophysiological rationale for automated seizure detection using machine learning algorithms.

However, the interpretation of EEG recordings presents several practical challenges. A typical continuous EEG recording from an epilepsy monitoring unit may span 24 to 72 hours, generating many gigabytes of data. Trained neurophysiologists must review this data manually, which is extremely time-consuming and subject to fatigue and inter-rater variability. Studies have shown that inter-rater agreement for seizure annotation from EEG recordings is often below 80%, reflecting the inherent ambiguity in visual EEG interpretation.

1.3 Machine Learning for EEG Analysis

The advent of deep learning has catalysed a revolution in automated EEG analysis. Convolutional Neural Networks (CNNs) have demonstrated remarkable ability to learn hierarchical spectrotemporal features from raw or minimally processed EEG signals without requiring hand-engineered feature extraction. Recurrent architectures such as Long Short-Term Memory (LSTM) networks have extended this to temporal modelling, capturing long-range dependencies within EEG sequences. Transformer models, originally developed for natural language processing, have been adapted for EEG through self-supervised pre-training approaches such as BENDR.

Despite these advances, several fundamental limitations persist in the current landscape of EEG-based seizure detection.

1.3.1 Inter-Subject Variability

EEG signals are highly patient-specific. Differences in skull thickness, scalp conductivity, cortical folding patterns, and the spatial origin of the epileptic focus mean that a model trained on one set of patients may generalise poorly to new, unseen patients. This challenge — known as the domain shift or distribution mismatch problem — severely limits the clinical

deployability of models that achieve high performance in within-subject evaluations but fail in cross-subject settings.

1.3.2 Class Imbalance

In real-world EEG recordings, seizure events constitute a tiny fraction of the total recording time. In the CHB-MIT Pediatric EEG dataset, for example, seizure windows represent approximately 0.5% of all available EEG windows. A naive classifier that predicts every window as “normal” would achieve 99.5% accuracy while entirely failing at its clinical purpose. Addressing this extreme class imbalance is therefore a critical preprocessing step.

1.3.3 Lack of Interpretability

Modern deep learning models are predominantly black boxes. Clinicians cannot adopt a system whose decision-making process they cannot audit or understand. The field of Explainable Artificial Intelligence (XAI) has produced powerful post-hoc attribution methods — notably SHAP and LIME — that can identify which input features most strongly influence a model’s predictions. However, fewer than 15% of published deep learning models for EEG analysis include post-hoc attribution analysis.

1.3.4 Spatial Relationships Between Channels

Most deep learning architectures treat EEG channels independently or concatenate them without explicitly modelling the spatial relationships between electrode positions. In reality, functional connectivity between electrode pairs carries rich discriminative information. Graph Neural Networks (GNNs) offer a principled framework for encoding these relationships, treating each electrode as a node and inter-electrode functional connectivity as a weighted edge.

1.4 Motivation for the Proposed Work

The convergence of these challenges motivates the development of a unified framework that simultaneously addresses cross-patient generalisation, spatial graph-based modelling, class imbalance, and post-hoc interpretability. This project presents the Domain-Adapted Graph

Representation Learning (DA-GRL) framework, which integrates all four capabilities within a single, end-to-end trainable architecture.

The DA-GRL framework is evaluated on the CHB-MIT Pediatric Scalp EEG Database — one of the most widely cited public benchmarks for seizure detection — using a rigorous experimental protocol that includes four architectural variants, a comprehensive preprocessing pipeline, and extensive XAI analysis.

1.5 Objectives of the Project

The specific objectives of this project are as follows:

1. To design and implement a comprehensive EEG preprocessing pipeline for the CHB-MIT dataset, including bandpass filtering, notch filtering, z-score normalisation, wavelet decomposition, and class balancing via SMOTE.
2. To develop a graph representation learning module that models inter-electrode functional connectivity as a learnable graph structure, enabling the spatial topology of the 10–20 electrode array to be explicitly exploited.
3. To integrate a gradient reversal layer-based domain adaptation module that enforces patient-invariant feature representations, thereby improving cross-patient generalisation.
4. To benchmark four architectural variants (SCT, GRL, DA-SCT, DA-GRL) on the CHB-MIT dataset and demonstrate the superiority of the proposed DA-GRL model.
5. To provide clinician-interpretable explanations through SHAP (GradientExplainer), LIME, temporal SHAP heatmaps, and channel importance rankings.
6. To identify the most clinically relevant EEG electrode pairs and seizure-sensitive temporal windows within each prediction window.

1.6 Scope and Significance

The proposed work makes several distinct contributions to the existing literature. First, it is one of the few studies to combine graph-based spatial modelling with adversarial domain

adaptation within a single framework for EEG-based seizure detection. Second, it provides a comprehensive XAI analysis using both SHAP and LIME, with a quantitative comparison of the two methods via Jaccard similarity and channel ranking correlation. Third, the temporal SHAP heatmap analysis reveals the specific time windows within a 4-second EEG segment where the model is most sensitive to seizure activity, providing a level of temporal granularity not available in prior work.

The practical significance of this work extends beyond benchmark performance metrics. A system that achieves 100% recall (zero missed seizures), provides transparent explanations grounded in established neurophysiology, and demonstrates robustness across diverse patients represents a meaningful step toward clinical deployment in epilepsy monitoring units.

1.7 Organisation of the Thesis

The remainder of this thesis is organised as follows. Chapter 2 presents a comprehensive survey of the relevant literature spanning graph neural networks for EEG analysis, domain adaptation methods, and XAI techniques. Chapter 3 articulates the problem statement. Chapter 4 describes the CHB-MIT Pediatric EEG Dataset in detail. Chapter 5 presents the EEG preprocessing methodology. Chapter 6 covers the feature engineering pipeline. Chapter 7 describes the proposed DA-GRL architecture. Chapter 8 reports the experimental results and model comparisons. Chapter 9 presents the explainable AI analysis. Chapter 10 concludes with a summary of findings and directions for future work.

CHAPTER 2

LITERATURE SURVEY

2.1 Introduction

Automated seizure detection using EEG signals has evolved through several algorithmic generations over the past three decades. This chapter presents a structured survey organised thematically: classical signal processing, deep CNNs, transformers, graph neural networks, domain adaptation, and explainable AI. Special attention is given to works addressing inter-subject variability, class imbalance, and interpretability — the three central challenges motivating this work.

2.2 Classical Signal Processing Approaches

Early automated seizure detection relied on hand-engineered features from spectral, temporal, and statistical properties of EEG. Subasi and Gursoy (2010) used Discrete Wavelet Transform (DWT) coefficients with a Support Vector Machine (SVM), achieving above 94% accuracy on a private dataset. Shoeb and Gutttag (2010) performed one of the first systematic studies on the CHB-MIT dataset using patient-specific SVMs with frequency-domain features. While competitive in sensitivity, the need for labelled seizure data from each target patient before deployment is a major practical limitation.

Bhattacharyya et al. (2017) applied multivariate empirical mode decomposition (MEMD) with instantaneous energy, reporting sensitivity 97.8% and specificity 98.1%. However, the computational complexity of MEMD limits real-time applicability. The key drawback of classical approaches is that hand-engineered features may be suboptimal and cannot adapt to data.

2.3 Convolutional Neural Network Approaches

Deep CNNs shifted the paradigm from hand-engineered to learned features. Acharya et al. (2018) used a 13-layer CNN on raw EEG segments, achieving 88.67% accuracy for five EEG states, proving end-to-end learning feasibility. Mekruksavanich et al. (2025) proposed an

enhanced hybrid CNN with squeeze-and-excitation attention, reporting sensitivity $\geq 97\%$ and specificity $\geq 98\%$ across multiple benchmarks. The attention mechanism allows the model to emphasise informative frequency bands and channels.

Golmohammadi et al. (2019) combined CNN and LSTM layers on the TUH EEG Corpus, capturing multi-scale temporal dependencies. Despite these advances, CNNs treat channels independently or as a 2D grid without explicit spatial encoding, missing the known electrode topology.

2.4 Transformer and Self-Supervised Approaches

Transformers use multi-head self-attention to capture long-range dependencies without the sequential bottleneck of RNNs. Kostas et al. (2021) introduced BENDR, a BERT-inspired contrastive self-supervised pre-training framework for EEG. Pre-trained on large unlabelled EEG, BENDR fine-tuned well for seizure classification with limited labels. Eldele et al. (2021) proposed TS-TCC, a temporal and contextual contrasting framework, producing robust representations.

Transformers are computationally expensive (attention scales quadratically) and still lack an inductive bias for electrode spatial relationships.

2.5 Graph Neural Network Approaches

GNNs are a natural fit for EEG because the electrode array defines a structured graph, and functional connectivity carries seizure-related information. Zhao et al. (2021) applied a linear graph convolution network (GCN) with focal loss on CHB-MIT, achieving AUC ≥ 0.98 . Li et al. (2022) extended to dynamic functional connectivity graphs using mutual information, reaching F1=0.961. Wang et al. (2022) incorporated phase-synchronisation (phase-locking value) into a spatiotemporal graph attention network, reporting AUC=0.982.

Zhang et al. (2022) used hypergraph neural networks for emotion recognition, modelling high-order relationships. However, most GNN-based seizure detection models do not address inter-subject variability nor provide explanations.

2.6 Domain Adaptation for EEG

Domain adaptation methods learn representations invariant to patient-specific shifts. Ganin et al. (2016) introduced the domain-adversarial neural network (DANN) with a gradient reversal layer (GRL). Jemal et al. (2024) applied adversarial alignment to cross-subject seizure prediction on CHB-MIT, improving AUC by up to 6 percentage points. Peng et al. (2022) combined GRL with Riemannian manifold alignment, reducing cross-subject error by 8%. Cui et al. (2023) proposed joint-probability-discrepancy minimisation, achieving 91.3% accuracy in leave-one-subject-out evaluation.

These methods improve generalisation but are rarely combined with graph-based spatial modelling or XAI.

2.7 Explainable AI in Biomedical Signal Processing

Lundberg and Lee (2017) introduced SHAP (SHapley Additive exPlanations), a game-theoretic framework providing global and local attributions. Ribeiro et al. (2016) proposed LIME, a local perturbation-based surrogate model. Salih et al. (2025) reviewed XAI in biomedical signal processing, noting that fewer than 15% of deep learning EEG models include post-hoc attribution. Giudice et al. (2022) applied SHAP to an EEG-based BCI, identifying theta/alpha features from frontal electrodes as most influential.

2.8 Research Gap Analysis

The literature survey reveals a clear gap: no existing system simultaneously provides robust cross-patient generalisation via domain adaptation, graph-based spatial modelling, and full SHAP-plus-LIME explainability validated against known neurophysiology on the CHB-MIT dataset. The proposed DA-GRL framework addresses this gap comprehensively.

CHAPTER 3

PROBLEM STATEMENT

3.1 Clinical Problem

Epilepsy management is challenged by (1) the enormous volume of EEG data from continuous monitoring, which precludes timely manual review, and (2) the absence of robust, interpretable automated seizure detection tools that generalise across diverse patients. Current systems often degrade dramatically on new patients (e.g., from 98% to 70% sensitivity) and are black boxes, providing no actionable information to clinicians.

3.2 Technical Problem Formulation

Let the EEG dataset consist of windowed segments with 18 channels and 1024 time samples (4 seconds at 256 Hz). Each segment has a binary label (seizure/normal) and a patient (domain) label. The domain shift problem arises because the distribution of EEG varies substantially across patients.

Technical objectives:

1. Learn a function that accurately predicts seizure probability.
2. Learn patient-invariant feature representations.
3. Explicitly model functional connectivity between electrodes via a graph.
4. Provide post-hoc attribution scores for each feature (channel $\times$ time).

3.3 Challenges Identified

3.3.1 Severe Class Imbalance

Raw CHB-MIT has a normal-to-seizure ratio of about 200:1. Standard cross-entropy training collapses to always predicting normal.

3.3.2 Inter-Subject EEG Variability

EEG morphology, amplitude, and seizure semiology vary widely across individuals due to differences in focus location, propagation pathways, and anatomy.

3.3.3 Missing Spatial Inductive Bias

CNNs and LSTMs process channels as sequences or 2D arrays without explicit electrode layout information, forcing the model to learn spatial relationships from scarce labelled data.

3.3.4 Black-Box Decision Making

Without explanations, clinicians cannot audit predictions, distinguish true seizures from artefacts, or take responsibility for AI-assisted decisions.

3.4 Proposed Solution Overview

The DA-GRL framework addresses each challenge:

- SMOTE oversampling + class-weighted loss → class imbalance.
- GRL-based domain adaptation → patient-invariant features.
- Graph convolution module → spatial relationships.
- SHAP + LIME → interpretability.

CHAPTER 4

DATASET DESCRIPTION

4.1 CHB-MIT Scalp EEG Database

The CHB-MIT Scalp EEG Database, hosted on PhysioNet, contains continuous scalp EEG from 24 paediatric patients (ages 1.5–22 years) with intractable epilepsy. Recordings were obtained after anti-seizure medication withdrawal to provoke seizures. The dataset uses the International 10–20 system with 18 bipolar channels. Sampling rate is 256 Hz (some at 512 Hz). Files are in European Data Format (EDF).

4.1.1 Patient Population

11 female, 14 male. Seizure counts per patient range from 3 to 40 (patient chb11 has the highest). Total seizure duration varies; patient chb14 has the longest cumulative seizure time.

4.1.2 Recording Protocol

The 18 standard channels: FP1-F7, F7-T7, T7-P7, P7-O1, FP1-F3, F3-C3, C3-P3, P3-O1, FP2-F4, F4-C4, C4-P4, P4-O2, FP2-F8, F8-T8, T8-P8, P8-O2, FZ-CZ, CZ-PZ. Bipolar montage suppresses common-mode noise.

4.1.3 Seizure Annotations

Each recording has a summary file with seizure onset and offset times annotated by a board-certified neurophysiologist. Total of 198 seizure events.

4.2 Dataset Statistics

Seizure duration ranges from 5 seconds to 300 seconds (mean 58.6 s, median 40 s). After non-overlapping 4-second windowing, the raw dataset contains 5,800 normal windows and 824 seizure windows (imbalance ratio 7:1). SMOTE balancing yields 5,800 per class, total 11,600 samples.

4.3 Data Access and Preprocessing Pipeline

Data is downloaded from PhysioNet, read using pyedflib, and preprocessed as described in Chapter 5.

CHAPTER 5

EEG PREPROCESSING METHODOLOGY

5.1 Introduction to EEG Preprocessing

Raw scalp EEG is contaminated by muscle artefacts, ocular artefacts, cardiac interference, electrode motion, and power line noise. A robust preprocessing pipeline is essential. This project implements five stages: bandpass filtering, notch filtering, z-score normalisation, wavelet decomposition, and SMOTE balancing.

5.2 Bandpass Filtering

A zero-phase 4th-order Butterworth bandpass filter (0.5–40 Hz) retains clinically relevant delta, theta, alpha, beta, and gamma bands while removing slow drift and high-frequency muscle noise. Zero-phase filtering (forward-backward) preserves event timing.

5.3 Notch Filtering

A narrow band-stop filter at 50 Hz removes power line interference without affecting nearby EEG frequencies.

5.4 Z-Score Normalisation

Each channel is normalised to zero mean and unit variance to remove amplitude differences across channels and patients, preventing bias toward high-amplitude channels.

5.5 Wavelet Decomposition

Discrete Wavelet Transform (DWT) with Daubechies-4 (db4) wavelet provides multi-resolution time-frequency decomposition. The db4 wavelet has compact support, preserving temporal resolution, and vanishing moments that capture spike-wave complexes.

5.6 Statistical Feature Extraction

For each window, we compute mean, standard deviation, skewness, kurtosis, and root mean square per channel. Seizure windows show elevated standard deviation and RMS.

5.7 Power Spectral Density Analysis

Welch's method estimates PSD. During seizures, delta band power increases approximately tenfold (from 440 to 4500 $\mu\text{V}^2/\text{Hz}$), a highly discriminative feature.

5.8 Channel Correlation Matrix

Pearson correlation matrix captures functional connectivity. It is used for exploratory analysis and to initialise the graph adjacency matrix.

5.9 SMOTE Class Balancing

SMOTE generates synthetic seizure samples by interpolating between existing seizure samples in feature space, reducing overfitting compared to simple duplication. After SMOTE, classes are balanced (5,800 each).

CHAPTER 6

FEATURE ENGINEERING

6.1 Overview

While DA-GRL operates on raw windows, feature engineering supports exploratory analysis and provides baselines. Features include time-domain (mean, std, skewness, kurtosis, RMS, zero-crossing rate), frequency-domain (band powers, spectral entropy), time-frequency (wavelet energy per level, wavelet entropy), and spatial (channel correlation).

6.2 Time-Domain Features

Seizure windows have significantly higher std and RMS. Kurtosis is elevated due to sharp spike-wave discharges.

6.3 Frequency-Domain Features

Relative band powers (delta, theta, alpha, beta, gamma) are clinically established. Spectral entropy decreases during seizures due to more regular, concentrated frequency distributions.

6.4 Time-Frequency Features

Wavelet energy at approximation level A5 (delta band) increases during seizures. Wavelet entropy measures order across scales; lower values indicate seizure activity.

6.5 Spatial Features

The correlation matrix reveals that adjacent channels have strong positive correlations. Seizures cause widespread increases in inter-channel correlation.

6.6 Feature Importance Analysis

An XGBoost classifier on the 90-dimensional statistical feature vector ranked RMS and std features for C3-P3, F3-C3, T8-P8, and F4-C4 as most important, consistent with later SHAP results.

CHAPTER 7

PROPOSED DA-GRL FRAMEWORK

7.1 System Overview

The DA-GRL framework has three main components: (i) temporal feature extraction (1D CNNs per channel), (ii) graph representation learning (graph convolution over electrode nodes), (iii) domain adaptation (gradient reversal layer + domain classifier). The architecture is end-to-end differentiable and jointly trained.

7.2 Input Representation

Input shape: (18 channels, 1024 time steps). Channels are processed independently initially.

7.3 Temporal Feature Extraction Module

Three 1D convolutional blocks (kernel size 3, stride 1), each with batch norm, ReLU, dropout (0.3). Channel dimensions: $1 \rightarrow 64 \rightarrow 128 \rightarrow 256$. Global average pooling yields a 256-dim feature vector per channel.

7.4 Graph Representation Learning Module

7.4.1 Graph Construction

Nodes = 18 channels. Edge weights initialised as positive Pearson correlations. Row-normalised adjacency matrix.

7.4.2 Graph Convolution Operation

Two graph convolution layers ($\text{in} \rightarrow 128$, $128 \rightarrow 128$) with ReLU. Global mean pooling over nodes gives a graph-level representation.

7.5 Domain Adaptation Module

7.5.1 Gradient Reversal Layer

The GRL acts as identity in forward pass but negates the gradient of the domain loss in backward pass, forcing features to be patient-invariant. The reversal coefficient is annealed from 0 to 1.

7.5.2 Domain Classifier

Two-layer MLP predicting patient identity from the graph-level representation (after GRL).

7.6 Classification Head and Loss Functions

7.6.1 Seizure Classification Head

Linear layer + sigmoid outputs seizure probability.

7.6.2 Combined Loss Function

Total loss = classification loss $\times$ domain loss. Classification loss is weighted binary cross-entropy (class weights to correct residual imbalance). Domain loss is multi-class cross-entropy.

7.7 Why DA-GRL Outperforms Conventional Models

Graph learning provides spatial inductive bias; domain adaptation enables cross-patient generalisation; joint end-to-end training optimises all components together. The combination yields zero missed seizures.

CHAPTER 8

EXPERIMENTAL RESULTS

8.1 Evaluation Protocol

80/10/10 stratified train/validation/test split. Metrics: accuracy, precision, recall, F1, AUC. Recall is clinically most critical (missed seizures are dangerous).

8.2 SCT Results

SCT (Spiking Convolutional Transformer) achieved accuracy 98.6%, precision 99.0%, recall 98.3%, F1 98.6%, AUC 0.999. Confusion matrix: 861 TN, 855 TP, 9 FP, 15 FN. The 15 false negatives are clinically concerning.

8.3 GRL Results

GRL (Graph Representation Learning) achieved accuracy 99.6%, precision 99.3%, recall 99.9%, F1 99.6%, AUC 1.000. Confusion matrix: 864 TN, 869 TP, 6 FP, 1 FN. The improvement over SCT demonstrates the benefit of graph-based spatial modelling.

8.4 DA-SCT Results

DA-SCT (Domain-Adapted SCT) achieved accuracy 98.3%, precision 99.2%, recall 97.4%, F1 98.3%, AUC 0.999. Adding domain adaptation to SCT reduced false negatives but also slightly lowered overall accuracy, suggesting training instability.

8.5 DA-GRL Results

DA-GRL (proposed) achieved accuracy 99.7%, precision 99.3%, recall 100%, F1 99.7%, AUC 1.000. Confusion matrix: 864 TN, 870 TP, 6 FP, 0 FN. Perfect recall — zero missed seizures — is the most clinically significant result. The false positive rate is only 0.69%.

8.6 Comprehensive Model Comparison

Table 8.1 summarises all models. DA-GRL outperforms all others. The ROC and precision-recall curves confirm near-perfect discrimination.

CHAPTER 9

EXPLAINABLE AI ANALYSIS

9.1 Motivation for Explainability

Clinicians need to understand, audit, and trust model decisions. Without explanations, verification is impossible.

9.2 SHAP Analysis

9.2.1 Channel Importance Ranking

SHAP values computed on 120 test samples (60 seizure, 60 normal) using GradientExplainer. Top channels: C3-P3, T8-P8, F3-C3, F4-C4, P7-O1. C3-P3 (central-parietal left) and T8-P8 (temporal-parietal right) align with neurophysiology of focal onset seizures.

9.2.2 Temporal SHAP Heatmap

High positive SHAP values concentrated around time steps 30–50 and 80–90 for C3-P3, indicating seizure-sensitive windows within the 4-second segment.

9.3 SHAP vs LIME Comparative Analysis

LIME identified FP1-F7 as top channel; Jaccard similarity for top-5 channels was 0.111. Partial disagreement is expected due to different mechanisms (gradient-based global vs perturbation-based local). SHAP is preferred for EEG because of theoretical guarantees, computational efficiency, global consistency, and neurophysiological plausibility.

9.4 Clinical Relevance

The SHAP results provide external validation: the model focuses on channels known to be involved in focal seizures, building clinician trust.

CHAPTER 10

CONCLUSION AND FUTURE SCOPE

10.1 Summary of Contributions

1. Comprehensive EEG preprocessing pipeline.
2. Graph representation learning module for functional connectivity.
3. GRL-based domain adaptation for patient-invariant features.
4. Benchmarking of four variants; DA-GRL achieves 99.7% accuracy, 100% recall.
5. XAI analysis (SHAP, LIME) identifying clinically relevant channels and time windows.

10.2 Key Findings

Perfect recall (zero missed seizures) is the most significant clinical finding. The model's explanations align with known neurophysiology, supporting trustworthiness.

10.3 Limitations

Single dataset (CHB-MIT), GPU dependence, static graph construction, retrospective evaluation only.

10.4 Future Scope

Multi-centre validation (TUH EEG Corpus), real-time implementation (sliding window, low latency), hospital integration and clinical trials, personalised online learning, federated learning for privacy preservation.

10.5 Final Remarks

DA-GRL meets the dual requirements of clinical-grade performance and interpretable reasoning, representing a meaningful step toward trustworthy AI-assisted neuro-diagnostics for the 50 million people with epilepsy worldwide.

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CHAPTER A

REPRESENTATIVE CODE LISTINGS

- A.1 EEG Preprocessing Pipeline**
- A.2 Gradient Reversal Layer Implementation**
- A.3 Graph Convolution Module**
- A.4 SHAP Channel Importance Computation**

CHAPTER B

HYPERPARAMETER CONFIGURATION

Table B.1 lists all hyperparameters used for DA-GRL training: learning rate $1e-3$, weight decay $1e-4$, batch size 64, epochs 30, dropout 0.3, GRL annealing schedule, two graph conv layers, three 1D conv blocks (64/128/256 filters), db4 wavelet, 5 DWT levels, SMOTE $k=5$, 80/10/10 split.